Phase 4 - organizer controls and abuse responses

Goal: An organizer can operate and observe a sale, while excessive buyer requests receive clear throttling without changing queue state. Estimated time: 4:00-4:45.

Build

1. Build /admin/events/[eventId] behind the organizer session. Add event open/close controls as supported, close/order, admit next batch, and expire due holds. Make repeated control actions safe using the Phase 2/3 idempotency rules.
2. Implement organizer-only GET /api/admin/events/:id/metrics. Read inventory, queue depth, admitted/reserved/sold counts, throttled/error counts, and recent decisions from the database. Display whether available + reserved + sold = total_inventory.
3. Add simple account-based rate limiting on excessive join/status requests, with IP as a secondary signal. Return 429, RATE_LIMITED, and retry guidance. A throttle response must not replace or delete the buyer's persisted queue state.
4. Emit audit reason codes for duplicate joins, throttling, expired admission/reservation, idempotency conflicts, and sold-out outcomes. Do not log secrets or full session cookies.
5. Add a safe demo reset path that creates a fresh run or clears only known seeded demo records. Preserve completed simulation runs and their results.
6. Check that admin controls cannot be invoked with a buyer session and metrics do not expose private account data through public routes.

Acceptance checks

- Dashboard counters reconcile with database rows and the inventory invariant indicator passes after transitions.
- Buyer sessions receive authorization errors on organizer routes.
- A flood of same-account requests produces visible 429 responses without losing the account's queue entry or position.
- Reset yields a clean demonstrable event without silently rewriting completed run evidence.

Handoff to Phase 5

Phase 5 needs stable admin metrics, audit reason codes, and a reset/isolation method for simulations. Record rate-limit thresholds, reset command, metric definitions, and known limits of the limiter.

Handoff record

Fill in after implementation: files/commits, commands, results, and open issues.